7-3 Logic Circuits

Point!

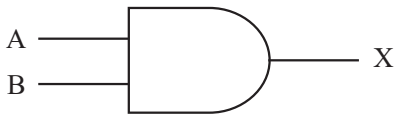
1 Logic Circuits

(1) ([1]**Logical operations**) : Operations performed using combinations of the numbers 0 and 1. In computers, "1" is processed as true and "0" as false.

(2) ([2]**Logic circuit**) : A circuit designed to perform logical operations.

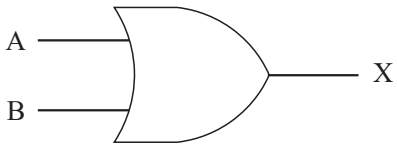
(3) ([3]**Truth table**) : A table that shows all possible combinations of inputs and outputs for a logic circuit.

(4) ([4]**AND gate (logical conjunction circuit)**) : A circuit that outputs 1 only when all inputs are 1.



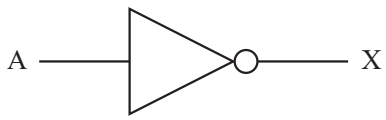
Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(5) ([5]**OR gate (logical disjunction circuit)**) : A circuit that outputs 1 if at least one of the inputs is 1.



Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

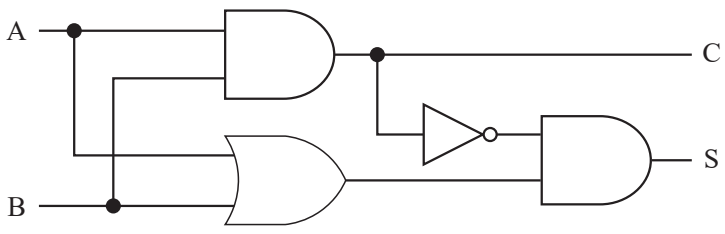
(6) ([6]**NOT circuit (inverter circuit)**) : A circuit that outputs the opposite result of the input.



Input	Output
A	X
0	1
1	0

2 Half Adder Circuit and Full Adder Circuit

(1) ([7]**Half adder circuit**) : A circuit representing the addition of single-digit numbers, composed of AND, OR, and NOT gates.



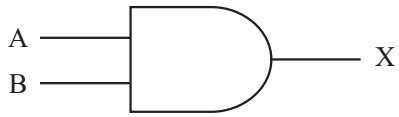
Input		Output	
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

(2) ([8]**Full adder circuit**) : A circuit that considers carry from the lower bit and carry to the higher bit.

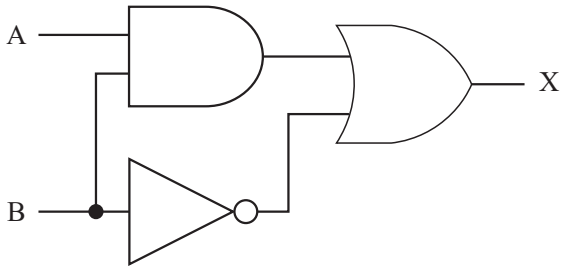
Warm Up

Answer the following questions.

(1) Name the following logic circuit and create a truth table.



(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



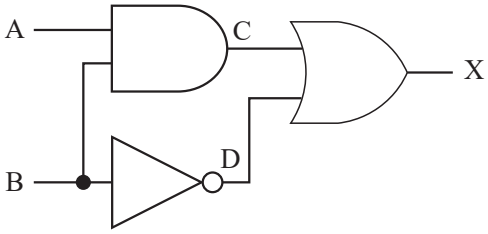
Explanation

(1) AND Circuit (logical conjunction circuit).

An AND circuit is a circuit that outputs 1 only when all inputs are 1. Therefore, the truth table is as follows.

Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(2) Create and analyze the truth table for each circuit.



A	B	C	D	X
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

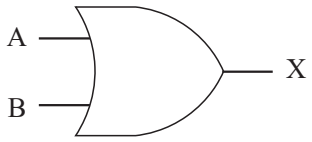
Therefore, the truth table is as follows.

Input		Output
A	B	X
0	0	1
0	1	0
1	0	1
1	1	1

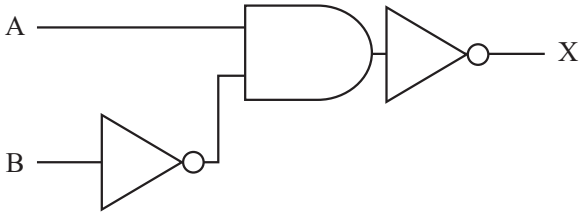
Try

Answer the following questions.

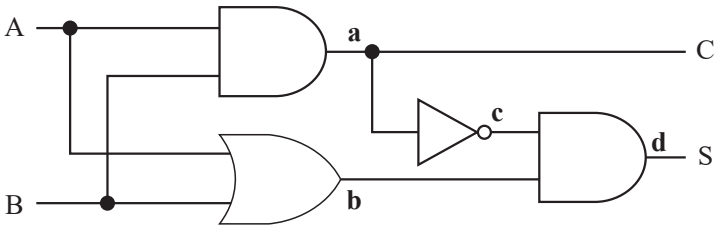
(1) Name the following logic circuit and create a truth table.



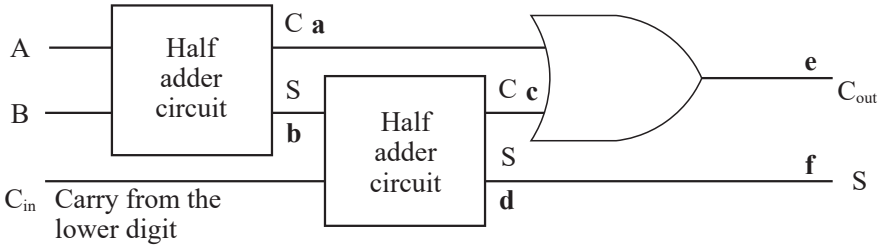
(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



(3) In the following half adder circuit, determine the values of each output **a** to **d** when input A is 0 and input B is 1.

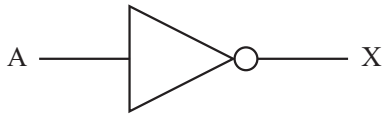


(4) In the following full adder circuit, determine the values of each output **a** to **f** when input A is 1, input B is 1, and input C_{in} is 0.

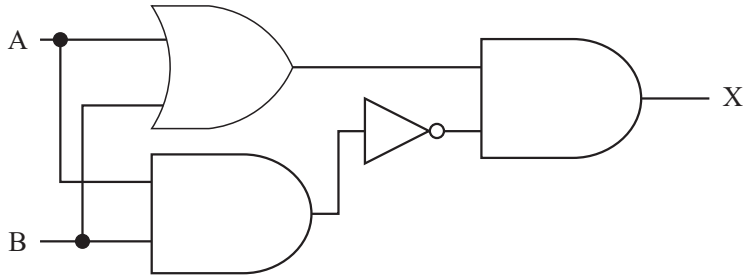


1 Answer the following questions.

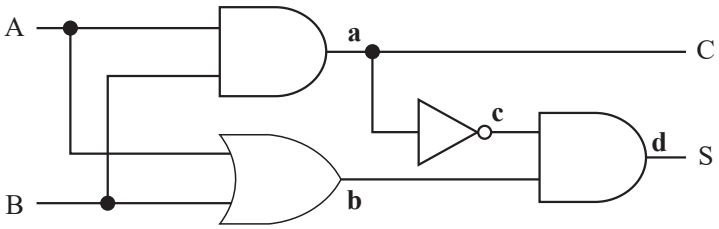
(1) Name the following logic circuit and create a truth table.



(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



(3) In the following half adder circuit, determine the values of each output **a** to **d** when input A is 1 and input B is 1.



(4) In the following full adder circuit, determine the values of each output **a** to **f** when input A is 1, input B is 0, and input C_{in} is 1.

